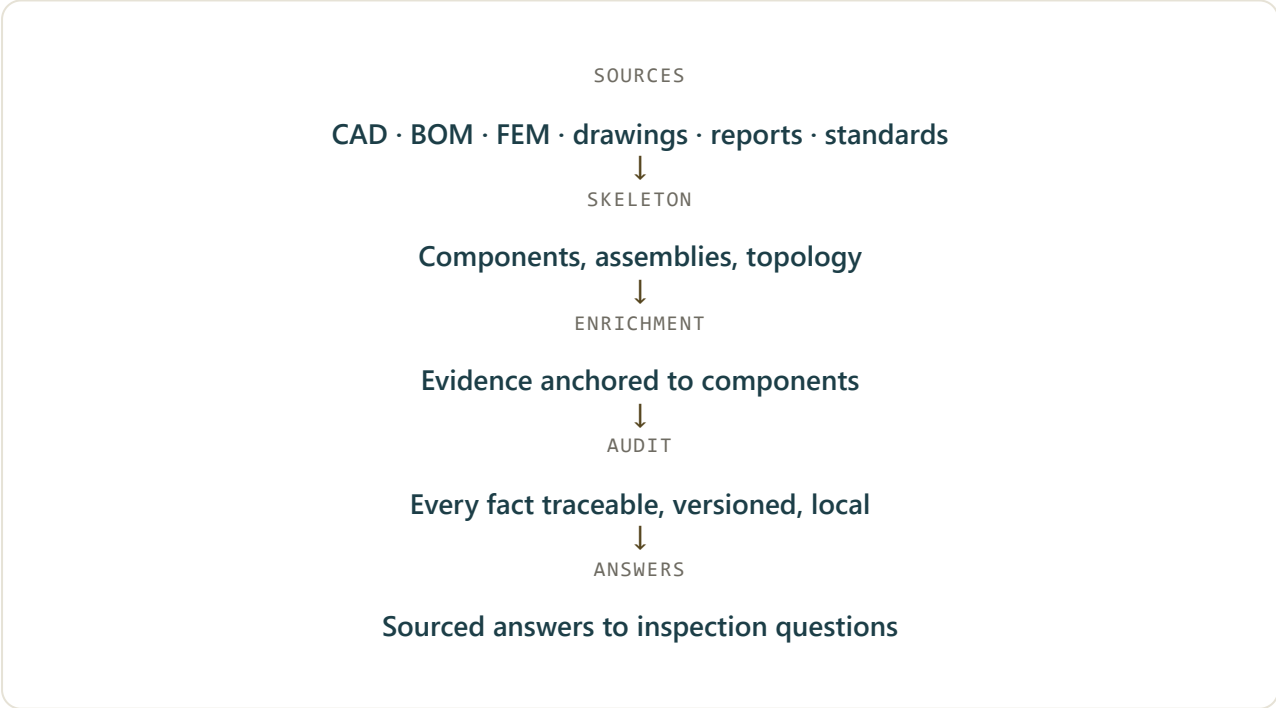
BAM · PACKAGE DESIGN ASSESSMENT · PROJECT AND TWO PHDS

A multi-source knowledge graph for the safety assessment of radioactive transport packages

Design-type assessment at BAM draws on scattered, multi-format engineering data. The **project** builds a shared data foundation (extraction and search) that is delivered on its own. Two PhDs build on top: **Mohamed's**, a component-centric, auditable knowledge graph, and **Alp's**, on evaluating how well AI agents reason and act compared to human experts. Three areas, three separate sets of work packages.

The idea in one line

Reliable structured data forms the skeleton. Reports, drawings and standards enrich it. Every statement stays traceable to its source.



Three areas, kept separate

It matters to keep these apart: one shared project layer plus two independent PhDs, each with its own work packages. Throughout: project is green (PW), Mohamed teal (AP), Alp plum (CW).

PROJECT · GREEN · PW

Shared foundation

Extraction, search and a benchmark for each (PW1-PW4), with more still to be scoped (multi-agent system, pipeline, its evaluation). Delivered on its own, independently of either PhD. Both rely on it.

MOHAMED · TEAL · AP

Knowledge graph

The component-centric, auditable knowledge graph (AP0-AP12). Builds on the foundation, for example enrichment uses the extraction, evaluation uses the search as a baseline.

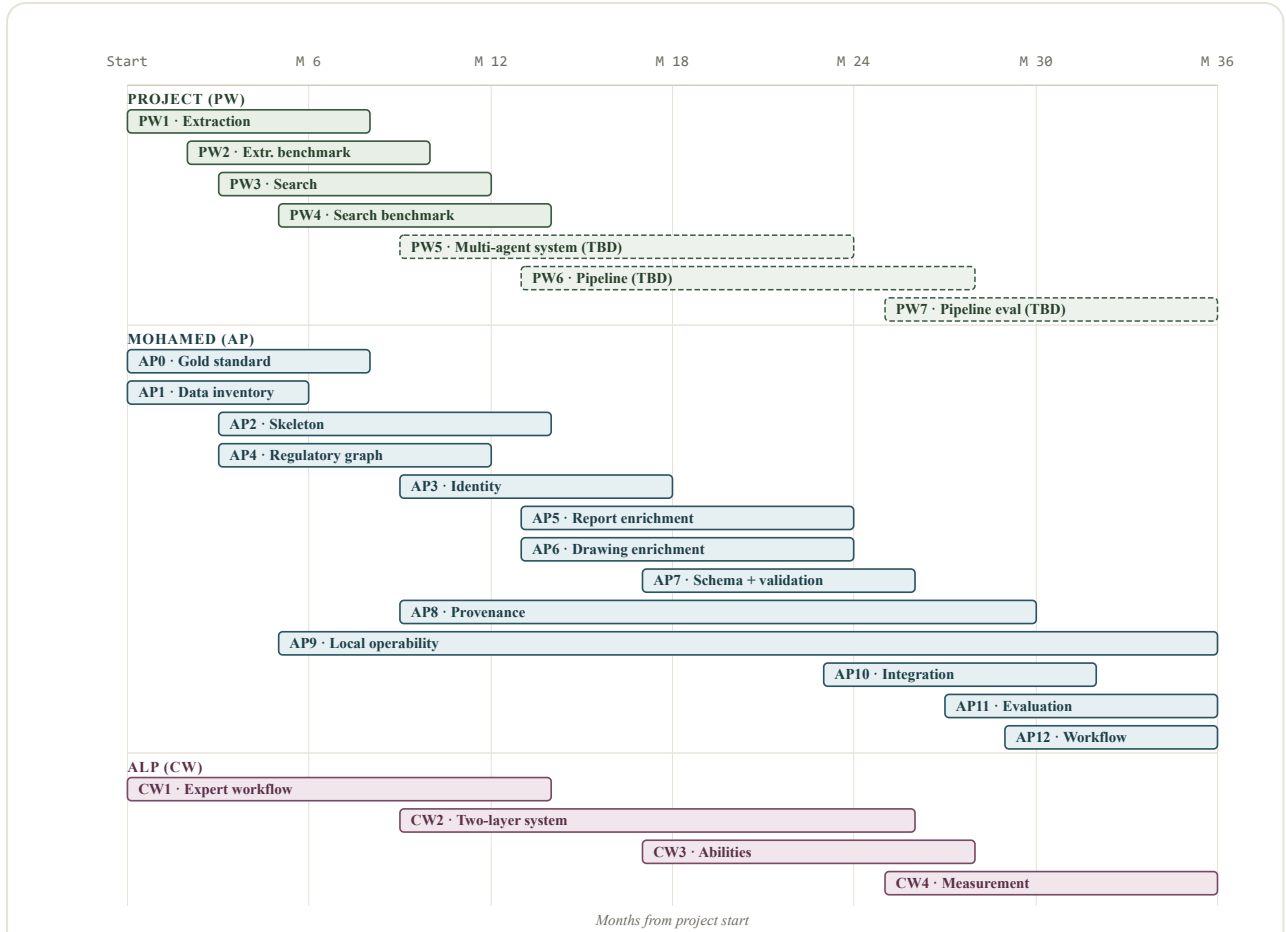
ALP · PLUM · CW

Agent evaluation

How well AI agents reason and act compared to human experts, ability by ability (CW1-CW4). Builds on the foundation, the measurement mostly towards the end.

Combined timeline

All three areas over time, kept visually separate. The project foundation (green) comes first and is delivered on its own. Mohamed (teal) and Alp (plum) build on top across three years; Alp's measurement sits late.



Project (PW), shared, delivered first Mohamed (AP) Alp (CW) dashed = not yet defined

Project foundation PROJECT · PW

Shared, project-wide, needed early. Also strong points for collaboration and theses, because the result is useful to the whole project independently of either PhD.

PW 1 Extraction from report PDFs

EXISTING, TO EXTEND

Extends the existing extraction pipeline so that text, tables, formulas, figures and layout are reliably obtained from the report PDFs.

PRODUCES

Extended extraction pipeline with structured output: text, tables, formulas, layout.

WHERE YOU HELP

Master thesis or domain input on tables, formulas and what good extraction looks like.

PW 2 Benchmark for extraction

BENCHMARK

Gold standard and metrics for the hard elements: tables, formulas and layout, so extraction quality is measurable per element type.

PRODUCES

Annotated extraction benchmark, metrics (precision, recall, table and formula accuracy).

WHERE YOU HELP

Domain experts annotating reference cases; methods groups on metrics.

PW 3 Search and retrieval

MASTER THESIS

Searchability through chunking, embedding and retrieval. Deliberately complex retrieval pipelines (hybrid, re-ranking, multi-stage), not just naive RAG.

PRODUCES

Configurable search pipeline: chunking, embeddings, retrieval, re-ranking, query interface.

WHERE YOU HELP

NLP or IR thesis; IT on infrastructure and model hosting.

PW 4 Benchmark for search

BENCHMARK

Gold standard and retrieval metrics to compare the search pipelines from PW3 with each other.

PRODUCES

Annotated search benchmark, metrics (Recall@k, MRR, nDCG), comparison of variants.

WHERE YOU HELP

Domain experts on relevance judgments; methods groups on evaluation.

Further project packages (not yet defined)

The project is more than extraction and search. These are placeholders for work the project will need but that is not scoped yet.

PW 5 Multi-agent system

TO BE DEFINED

The project-level orchestration of the agents that produce the assessment. Scope and design still open. This is the project's shared agent layer, not Mohamed's KG-constructing agents (AP10) and not Alp's evaluation system (CW2).

PW 6 **Pipeline** TO BE DEFINED

Orchestrates the whole system (extraction, search, KG construction) into one runnable flow. Scope still open. This is the overarching orchestration, not the KG construction itself (that is Mohamed's AP10).

PW 7 **Pipeline evaluation** TO BE DEFINED

Technical end-to-end evaluation of the pipeline: does the whole thing work, and how well. This is distinct from Alp's evaluation (CW), which measures how the agents reason and act compared to human experts.

Mohamed: the methodological core PHD · AP

These five packages hold the method together and stay with Mohamed's PhD. Shown here for the overall picture, not for delegation.

AP 2	**Engineering skeleton (CAD, BOM, FEM)**	The reliable structured core of the graph: components, assemblies, topology, materials.
AP 3	**Component identity**	When two mentions from different sources mean the same component, and when they must stay separate.
AP 7	**Schema profile and validation**	Completeness and consistency checks against the regulatory target structure.
AP 10	**Integration and demonstrator**	The local end-to-end pipeline: sources in, auditable component-centric graph out.
AP 11	**Evaluation: manual vs. RAG vs. graph**	Whether the graph really beats manual inspection and simple retrieval (uses PW3 as baseline).

Mohamed: where your team could come in PHD · AP

These packages have a natural opening for collaboration. They are starting points, not assignments: where any of them touches your team's work or interest, there is a natural opening to shape it together.

AP 0 Gold standard to test the KG

DOMAIN VALIDATION

A reference the constructed knowledge graph is measured against: for selected application cases the correct target graph plus example questions with expected answers. Not about capturing the expert's review process (that is Alp's CW1), but about having a correct target to measure against. How the gold standard is built is still open.

PRODUCES

Reference annotation for selected application cases, test questions with expected answers, evaluation criteria.

WHERE YOU HELP

Domain validation that the reference graph is correct.

AP 1 Data inventory, privacy and interfaces

IT + DOMAIN

Clarifies which data really exists, in which formats, and how privacy and anonymization are handled.

PRODUCES

Inventory matrix for 3-5 cases, format list, privacy concept.

WHERE YOU HELP

IT on interfaces and storage, domain on which cases are representative.

AP 4 Regulatory and evidence graph

STANDARDS EXPERTS

Turns SSR-6 and SSG-66 requirements into a structured target form for completeness and consistency checks.

PRODUCES

Requirement sub-graph: evidence types, component classes, conditions, cross-references.

WHERE YOU HELP

Regulatory experts interpreting clauses and the evidence expected per component.

AP 5 Evidence enrichment from reports

MASTER THESIS

Takes the raw extraction from PW1 and anchors statements, load assumptions and evidence to the right component.

PRODUCES

Component-centric enrichment with claim-plus-evidence and source spans.

WHERE YOU HELP

NLP-oriented thesis; domain experts judging enrichment quality.

AP 6 Drawing enrichment

MASTER THESIS

Links dimensions, tolerances and material data from drawings to the right component.

PRODUCES

Vision and OCR prototype for selected drawing types plus an evaluation set.

WHERE YOU HELP

Computer-vision thesis; engineering selecting representative drawing types.

AP 8 Provenance and versioning

IT + METHODS

Keeps every statement traceable to source, document version and section across all processing steps.

PRODUCES

Provenance model, versioning strategy, audit log.

WHERE YOU HELP

IT and methods groups on storage mechanisms and audit trails.

AP 9 Local operability

IT INFRASTRUCTURE

Determines which local models, quantizations and hardware carry the pipeline without any cloud.

PRODUCES

Model recommendation with benchmarks, reproducible bench suite, local runtime profile.

WHERE YOU HELP

IT on infrastructure and model hosting; methods groups on benchmarking.

AP 12 Inspection workflow and human validation

OPTIONAL · INSPECTORS

Studies how inspectors are best involved in validation, conflict resolution and release.

PRODUCES

Workflow concept, validation mockup, study of effort and error rate.

WHERE YOU HELP

Inspectors providing time and feedback; an HCI-oriented thesis.

Alp: agent evaluation

PHD · CW

The second PhD in the project. Focus: not how well the agents extract or search, but how well they think and act compared to a human expert. Expert review is decomposed into cognitive abilities, and the system is measured ability by ability against a golden set.

CW 1 Expert-workflow extraction

DOMAIN CAPTURE

AI-assisted capture of how experts actually review safety reports (3-5 experts, about 200 unique tasks, 1000+ in total). Builds the basis for the golden set.

PRODUCES

Annotated expert-workflow corpus and the golden set.

WHERE YOU HELP

Inspectors contributing review work; domain experts validating tasks.

CW 2 Two-layer assessment system

CORE (ALP)

A pipeline that reasons step by step and acts on its written reasoning (each step a separate, stateless skill), plus an architecturally separate evaluation layer that checks provenance, correctness, completeness and reasoning. Alp evaluates agents in general, independent of the project's and Mohamed's KG agents.

PRODUCES

Two-layer agent system with reasoning kept separate from evaluation.

WHERE YOU HELP

Methods and AI groups on agent design and the evaluation layer.

CW 3 Categorise abilities

FRAMEWORK

Decompose the golden set into cognitive abilities using the EU Joint Research Centre framework, so that "what an expert does" becomes a structured list of abilities.

PRODUCES

Ability taxonomy mapped onto the golden set.

WHERE YOU HELP

Domain experts on what abilities a good review actually requires.

CW 4 Quantitative ability measurement

EVALUATION · LATE

Score the two-layer system ability by ability against the golden set: how well the agent thinks and acts versus a human expert, tracked across iterations. Hard to place in time, mostly towards the end.

PRODUCES

Per-ability scores, improvement tracking across iterations.

WHERE YOU HELP

Inspectors as the human reference; methods groups on scoring.

Data stays in the house

> **On-premise by design.** All processing runs on local BAM infrastructure. Application data never leaves the house, no commercial cloud models are used. Traceability to source and versioning are built into the method from the start (AP8), not added afterwards.
